

where wage is measured in dollars per hour, education and experience are in years, and $METRO = 1$ if the person lives in a metropolitan area. We have $N = 1000$ observations from 2013.

- $$WAGE_i = \beta_1 + \beta_2 EDUC_i + \beta_3 EXPER_i + \beta_4 METRO_i + \beta_5 FEMALE + e_i \quad (XR8.6b)$$

d. Following the regression in part (b) we carry out the White test for heteroskedasticity. The value of the test statistic is 78.82. What are the degrees of freedom of the test statistic? What is the 5% critical value for the test? What do you conclude?

$$\begin{cases} H_0: \sigma_M^2 = \sigma_F^2 \\ H_1: \sigma_M^2 \neq \sigma_F^2 \end{cases}$$

$$\left\{ \begin{array}{l} \hat{\sigma}_M^2 = \frac{SSE_M}{df_M} \approx 169.567 \\ \hat{\sigma}_F^2 = (12.024) \approx 144.577 \end{array} \right.$$

$$F = \frac{\hat{\sigma}_M^2}{\hat{\sigma}_E^2} = \frac{169.567}{144.577} \doteq 1.1728$$

$$F_{0.025, 573, 419} \approx 1.168$$

$\therefore F > F_{0.025, 573, 419}$, then reject H_0 .

$$NR^2 = 59.03$$

$$\chi^2_{(0.05, 4)} \approx 9.488$$

$\therefore NR^2 > \chi^2_{(10.05, 4)}$. then reject H_0 .

No, because the Married variables was not included in this regression.

So this test tell us nothing about marital status affects wage variation.

$$\begin{cases} H_0: \sigma_{\text{single}}^2 = \sigma_{\text{married}}^2 \\ H_1: \sigma_{\text{single}}^2 < \sigma_{\text{married}}^2 \end{cases}$$

$$df_{\text{single}} = 400 - 5 = 395, \quad df_{\text{married}} = 600 - 5 = 595$$

$$\begin{cases} \hat{\sigma}_{\text{single}}^2 = (56231.0382) / 395 \approx 142.3571 \\ \hat{\sigma}_{\text{married}}^2 = (100703.0471) / 595 \approx 169.2488 \end{cases}$$

$$F = \frac{\hat{\sigma}_{\text{married}}^2}{\hat{\sigma}_{\text{single}}^2} = \frac{169.2488}{142.7571} \approx 1.1889$$

$$F_{0.05, 395, 595} \approx 1.163$$

$\therefore F > F_{0.05, 95, 595}$, then reject H_0 .

1.1 $\begin{cases} \text{original variables} : \text{EDUC, EXPER, METRO, FEMALE} \\ \text{squared variables} : \text{EDUC}^2, \text{EXPER}^2 \\ \text{cross-products} : \text{EDUC} \times \text{EXPER} \quad \text{EDUC} \times \text{METRO} \quad \text{EDUC} \times \text{FEMALE} \\ \quad \quad \quad \text{EXPER} \times \text{METRO} \quad \text{EXPER} \times \text{FEMALE} \quad \text{METRO} \times \text{FEMALE} \end{cases}$

$$\therefore df = 12 \Rightarrow \chi^2_{(0.05, 12)} = 21.026$$

$$\therefore 78.82 > 21.026$$

$\therefore$ reject H_0 , heteroskedasticity exists.

- e. The OLS fitted model from part (b), with usual and robust standard errors, is

$$\widehat{WAGE} = -17.77 + 2.50EDUC + 0.23EXPER + 3.23METRO - 4.20FEMALE$$

(se)	(2.36)	(0.14)	(0.031)	(1.05)	(0.81)
(robse)	(2.50)	(0.16)	(0.029)	(0.84)	(0.80)

For which coefficients have interval estimates gotten narrower? For which coefficients have interval estimates gotten wider? Is there an inconsistency in the results?

- f. If we add *MARRIED* to the model in part (b), we find that its *t*-value using a White heteroskedasticity robust standard error is about 1.0. Does this conflict with, or is it compatible with, the result in (b) concerning heteroskedasticity? Explain.

e.

narrower : EXPER , METRO , FEMALE

wider : EDUC

No, there is no inconsistency. The mathematical formula for White's *rbse* allows them to be either larger or smaller than usual OLS *s.e.* in any given finite sample.

f. The results are completely compatible.

∴ (f) shows that marital status doesn't shift the average level of a person's wage.

(b) shows that marital status change the spread or dispersion of a person's wage. around that average.

∴ it is entirely possible.